Practice Set 9 Paper 1 – Basic Sciences in Urology

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Practice paper compiled from the question bank.

Q1. Describe in detail, anatomy of prostate with special reference to concepts in surgical management if localised CA prostate.

Type: LAQ (25 marks) (also answered in Practice Set 2 Paper 4, Q1)

"if localised" is read as "of localised" prostate cancer.

Introduction

- A 20-g fibromuscular gland around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**).
- RP (radical prostatectomy) aims at the "trifecta" of cancer control, continence and potency. Each anatomical step either secures a clear margin or saves a named structure (nerves, sphincter, anterior support).

Zonal anatomy (McNeal, 1981)

Zone	Share of glands	Surgical relevance
Peripheral (PZ)	About 70%	**70% of cancers**; posterolateral, so ECE (extracapsular extension) runs along the nerves
Central (CZ)	About 25%	Cone around the ejaculatory ducts; few cancers
Transition (TZ)	About 5%	BPH (benign prostatic hyperplasia); anterior cancers missed by DRE (digital rectal examination)
AFMS (anterior fibromuscular stroma)	No glands	Anterior tumours reach it; margin risk in posterior approaches

- There is **no true capsule**, so cancer breaching the gland is "extraprostatic extension".

Fascial planes and nerve sparing

Plane	Lies	Trade-off
Intrafascial	Between capsule and prostatic fascia	Most nerves kept; more positive margins
Interfascial	Between prostatic and levator fascia	**Standard** for organ-confined disease
Extrafascial	Outside levator and Denonvilliers fascia	Widest margin; ED (erectile dysfunction) likely

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral ECE is suspected.

Neurovascular bundle (NVB)

- Walsh and Donker (1982). The pelvic plexus (S2–S4 parasympathetic plus hypogastric sympathetic) gives the **cavernous nerves**. They run posterolaterally **3–5 mm from the capsule**, marked by capsular vessels.
- At the apex they lie at **5 and 7 o'clock**; most nerve injuries occur here.
- Nerve distribution is diffuse (posterolateral bundle, posterior "hammock", anterolateral fibres), which is the basis of high anterior release.
- Injury comes from incision, entrapment, **heat** and **traction**, so dissect athermally without traction.

Vessels and lymphatics

- Inferior vesical artery (urethral and capsular branches). Preserve **accessory pudendal arteries**; they may be the main penile supply.

- DVC (dorsal venous complex): CO_2 pneumoperitoneum tamponades it, so robotic blood loss is low.
- Lymph: obturator, external and internal iliac, common iliac, presacral.

Continence anatomy

Passive	Active
Smooth muscle of bladder neck and urethra; anterior support (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC)	Horseshoe striated sphincter at the apex, supplied by pudendal nerve branches

Concepts in surgical management of localised cancer

1. **Selection:** RP suits localised disease (mainly intermediate and selected high risk) with a life expectancy of more than 10 years; low risk is usually offered active surveillance.
2. **RARP (robot-assisted RP):** 3D, magnified view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal early function versus open; less blood loss and shorter stay. Surgeon matters more than platform.
3. **Risk-graded nerve sparing:** each side is graded from full (intrafascial) through partial to none, using grade, positive cores and mpMRI (multiparametric magnetic resonance imaging) capsular contact. **NeuroSAFE** (intraoperative frozen section of the posterolateral surface) aims to allow more nerve sparing without more positive margins.
4. **Anterior-support preservation:**
    - **Retzius-sparing** (Bocciardi, 2010): fully posterior through the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins, so avoid for anterior tumours.
    - **Hood technique** (Tewari, 2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC.
5. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length and divide sharply without cautery. Posterior (Rocco) reconstruction may aid early continence.
6. **PLND (pelvic lymph node dissection) templates:**

Template	Nodes	Drainage captured
Limited	Obturator	Least
Standard	+ External iliac	About 38%
Extended	+ Internal iliac	About 63%
+ Common iliac	+ Common iliac	About 75%

- Extended PLND when nodal risk exceeds 5% (EAU [European Association of Urology]). It stages but does not improve survival; lymphocele is the commonest complication.

7. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy. Spare sphincter, NVB and rectum.

- **Also know:** single-port robot (extraperitoneal); PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; ERAS (enhanced recovery after surgery).

Diagram to draw: axial prostate showing the three fasciae, the NVB at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

Key points to remember:

- PZ gives 70% of cancers; no true capsule.
- Interfascial plane is the standard nerve-sparing plane; apex is the commonest margin and nerve-injury site.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

Q2. Describe anatomy of pelvic floor in females. Discuss fertility preserving treatment options for procidentia

Type: LAQ (25 marks) (also answered in Practice Set 2 Paper 1, Q1)

"Fertility preserving" means uterus-preserving treatment for a woman who wants future pregnancy.

A. Anatomy of the female pelvic floor

Support comes from the **levator ani**, the **endopelvic fascia and ligaments**, and the **perineal membrane and perineal body**, anchored to the bony pelvis.

1. Pelvic diaphragm

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction	Anorectal angle
Pubococcygeus	Pubis and arcus tendineus to viscera and anococcygeal raphe	Slings urethra and vagina
Iliococcygeus	Arcus tendineus and ischial spine to coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine to sacrum and coccyx	Lies on sacrospinous ligament

- Nerve to levator ani **(S3–S5)**; **pudendal nerve (S2–S4)** to sphincters and perineal muscles.
- Tonic contraction closes the **urogenital hiatus**; the uterus and upper vagina rest on the levator plate.

2. Perineal membrane and perineal body

- The membrane spans the pubic arch and is pierced by urethra and vagina; the striated urogenital sphincter lies above it. The perineal body anchors the distal vagina.

3. Ligaments and surgical anchors

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix). The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix.
- Pubocervical fascia attaches to the **ATFP (arcus tendineus fasciae pelvis)**, pubis to ischial spine.
- **Sacrospinous ligament (SSL)**, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine.
- **Sacral promontory**: left common iliac vein and right ureter each about 2.7 cm away.

4. DeLancey levels (1992)

Level	Structures	Loss produces
I suspension	Cardinal–uterosacral complex	**Uterine or vault prolapse**, enterocele
II attachment	Paravaginal attachment to ATFP	Anterior or posterior wall prolapse
III fusion	Perineal membrane, perineal body	Urethral hypermobility, distal rectocele

5. Why prolapse occurs

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens and the ligaments carry the whole load and stretch. Other risks: vaginal birth, forceps, age, obesity, constipation, collagen disorders.

Diagram to draw: sagittal female pelvis with levator plate and DeLancey levels I–III.

B. Procidentia: definition and assessment

- **Procidentia**: complete uterine (or vault) prolapse outside the hymen, **POP-Q (pelvic organ prolapse quantification) stage 4**.
- Assess all compartments, occult SUI (stress urinary incontinence) with prolapse reduced, PVR (post-void residual), hydronephrosis (ureteric kinking at the hiatus), cervical length and ulceration.
- Abnormal bleeding or cervical pathology needs gynaecological evaluation first, because it may argue against keeping the uterus.

C. Fertility-preserving options

Principle: restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

1. Conservative (first line)

- PFMT (pelvic floor muscle training), weight loss, treat constipation.
- **Pessary**: space-filling type (Gellhorn, cube) for procidentia; removed for conception; regular review for discharge and erosion.

2. Uterus-preserving surgery

Operation	Technique	Fertility points
Sacrospinous hysteropexy (vaginal)	Cervix sutured into the SSL **1.5–3 cm medial to the ischial spine**	Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); later data suggest superiority. Buttock pain, pudendal injury
Uterosacral hysteropexy (vaginal or laparoscopic)	Shorten and reattach uterosacral ligaments to cervix	No mesh; cystoscopy for ureteric kinking
Sacrohysteropexy (open, laparoscopic, robotic)	Mesh from posterior cervix to anterior longitudinal ligament over the promontory	Most durable apical support; later delivery usually by caesarean
Manchester (Fothergill)	Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy	Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted

- **Also know:** Indian cervicopexy slings for young nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

3. Choosing

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Procidentia, wants pregnancy -> PFMT + pessary
  -> fails -> sacrospinous hysteropexy or
     laparoscopic sacrohysteropexy (avoid Manchester)
  -> recurrence after childbearing -> hysterectomy
     + vault suspension or sacrocolpopexy
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D. Complications and follow-up

- Recurrence (mostly posterior), de novo SUI, mesh exposure (3.4–5.4% vaginal mesh exposure in meta-analyses), ureteric kinking. The retained uterus needs cervical screening.

Key points to remember:

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey level I (uterosacral–cardinal) failure causes uterine prolapse.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

Q3a. Describe contribution of a. Prof. Herald Hopkins b. Dr Charles Huggins c. Prof. Roy Caine d. Dr D D Gaue

Type: SAQ (10 marks)

Names read as Prof. Harold Hopkins, Dr Charles Huggins, Prof. Sir Roy Calne and Dr D. D. Gaur.

a. Prof. Harold Hopkins (British physicist)

- **Fibreoptics (1954, with Kapany):** coherent bundles of glass fibres transmit an image by total internal reflection (Tyndall's principle). This made flexible endoscopes possible.
- **Rod-lens system (1966):** the old scope used thin glass lenses separated by long air spaces. Hopkins replaced them with **long glass rods separated by short air gaps**. Result: more light transmission, a brighter and wider image, and a **smaller calibre**. Developed commercially with Karl Storz.
- It underpins all modern rigid endoscopes and enabled the first rigid ureteroscope (1979).

b. Dr Charles Huggins (urologist, University of Chicago)

- **1941, with Hodges:** showed that prostate cancer is **androgen-dependent**. Castration (or oestrogen) relieved far-advanced disease; injected androgens aggravated it.
- Used serum **acid phosphatase** to monitor response, an early tumour marker.
- **1945:** bilateral adrenalectomy for castration-resistant disease (modest, brief responses).
- **Nobel Prize in Physiology or Medicine, 1966**: the basis of ADT (androgen deprivation therapy) and of all hormonal cancer therapy.

c. Prof. Sir Roy Calne (Cambridge transplant surgeon)

- Showed in dogs that **6-mercaptopurine and then azathioprine** prolong renal allograft survival; azathioprine with steroids became standard immunosuppression.
- Performed the **first liver transplant in the UK (1968)**.
- **First clinical use of ciclosporin** in kidney transplantation (late 1970s), the calcineurin-inhibitor era; acute rejection fell markedly.

d. Dr D. D. Gaur (Mumbai, India)

- **1992: balloon dissection of the retroperitoneum.** A size 7 surgeon's glove tied on an 8-Fr red rubber catheter, inflated inside the retroperitoneum, creates a working space.
- This made **retroperitoneoscopy** practical and cheap; used for ureterolithotomy, nephrectomy and other renal surgery.

Diagram to draw: conventional lens scope versus Hopkins rod-lens scope (glass rods with short air gaps).

> **Key points to remember:**
>
> - Hopkins: rod-lens system 1966; fibreoptics 1954.
> - Huggins and Hodges 1941: androgen dependence; Nobel 1966.
> - Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
> - Gaur 1992: glove-balloon retroperitoneoscopy.

Q3b. EMG

Type: SAQ (10 marks)

EMG is taken as urodynamic electromyography of the striated urethral/anal sphincter and pelvic floor.

Definition and principle

- EMG (electromyography) records the electrical potentials of depolarising striated muscle membranes, here the **external urethral sphincter and pelvic floor**, during filling and voiding.
- It shows **coordination** between detrusor and sphincter, not sphincter strength. The bladder neck (smooth muscle) is invisible to EMG.
- External anal sphincter activity mirrors the urethral sphincter, except in some neurological diseases.

Electrodes

Surface (patch)	Concentric needle
Perianal patches at 3 and 9 o'clock (ICS [International Continence Society] standard: two skin electrodes plus reference)	Needle into the sphincter (girls beside the meatus; boys transperineal)
Comfortable; poorer signal; movement artefact	Best signal; painful; dislodges
Coordination only	Also motor unit analysis: acute versus chronic denervation

Normal pattern

- Filling: activity **rises gradually** (recruitment, guarding reflex).
- Voiding: EMG **falls silent before** detrusor pressure rises and stays silent until flow ends.
- **Lag time** (uroflow with patch EMG in children): relaxation to flow 2–6 s is normal; <2 s suggests DO (detrusor overactivity); >6 s suggests primary bladder neck dysfunction.

Abnormal patterns

Finding	Meaning
DSD (detrusor–sphincter dyssynergia)	Sphincter contracts during detrusor contraction; needs a **suprasacral** cord lesion (SCI [spinal cord injury], MS [multiple sclerosis]). Blaivas types 1–3
Dysfunctional voiding	Same tracing without neurological lesion (learned)
Pseudodyssynergia	EMG rise from straining or trying to suppress a contraction
Denervation potentials	Sacral cord or cauda equina lesion; MSA (multiple system atrophy)
Complex repetitive discharges, decelerating bursts	**Fowler syndrome** (retention in young women); predicts SNM (sacral neuromodulation) response

Clinical uses

- Neurogenic bladder (spina bifida, SCI); voiding dysfunction in children; female retention; before sphincter surgery.
- **Limitation:** surface EMG misdiagnoses some dysfunctional voiders; **VUDS (videourodynamics)** is more accurate.

Diagram to draw: tracing of Pdet, flow and EMG: EMG silent during normal voiding versus active in DSD.

Key points to remember:

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

Q3c. Secondary malignancies in Uro

Type: SAQ (10 marks) (also answered in Practice Set 3 Paper 1, Q3f)

Read as second cancers caused by treatment (radiotherapy, chemotherapy, urinary diversion, immunosuppression).

Definition

- A **second (treatment-related) malignancy** is a new, histologically distinct cancer caused by treatment of a first disease. It appears after a latency of years, often **in the radiation field**.

Causes and risks

Cause	Second cancer	Key facts
Pelvic radiotherapy	**Bladder**, rectal cancer, sarcoma (in-field)	Bladder RR (relative risk) 2–4 after cervical EBRT (external beam radiotherapy), 1.5–2 after prostate EBRT; little with IMRT (intensity-modulated radiotherapy). Latency 15–30 years
Cyclophosphamide	**Bladder cancer**, UTUC (upper tract urothelial carcinoma)	Only chemotherapy proven to cause bladder cancer (acrolein, phosphoramide mustard); about 4.5-fold, dose-dependent
Urinary diversion	Adenocarcinoma at the uro-intestinal junction	Ureterosigmoidostomy 6–29% after 10–20 years; cystoplasty 1.6%; ileal conduit 0.02%
Transplant immunosuppression	Skin cancer, PTLD (post-transplant lymphoproliferative disorder), RCC (renal cell carcinoma), bladder	RCC mostly in native kidneys; bladder about 3-fold
Testicular GCT (germ cell tumour) treatment	Solid tumours, leukaemia	Excess after cisplatin or radiotherapy, highest after both; etoposide leukaemia

Mechanisms

- Radiation and alkylating drugs: mutagenic DNA damage.
- Immunosuppression: lost immune surveillance, oncogenic viruses (EBV [Epstein–Barr virus], HPV [human papillomavirus]).
- Diversion: urine–faeces contact at the anastomosis.

Prevention and surveillance

- IMRT, smaller fields; **mesna** with cyclophosphamide.
- **Haematuria after pelvic radiotherapy or cyclophosphamide: always cystoscopy.**
- Endoscopy from 5 years after ureterosigmoidostomy; augmentation surveillance from about 10 years; excise a defunctioned ureterosigmoid anastomosis.
- Testis cancer: fewer treatment modalities; low-dose CT (computed tomography) in surveillance.
- **Also know:** paediatric tumours (Wilms 1% at 10 years); pioglitazone, aristolochic acid, phenacetin.

Diagram to draw: ureterosigmoid anastomosis with adenocarcinoma at the junction.

Key points to remember:

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

Q3d. Hematospermia

Type: SAQ (10 marks) (also answered in 2025-Winter Paper 1, Q3c)

Definition

- Blood in the ejaculate. Usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.
- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified. **Pseudo-haematospermia:** partner's blood (condom test) or haematuria.

Causes

Group	Examples
Iatrogenic (commonest identifiable)	Prostate biopsy (about 10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy
Inflammation and infection	Prostatitis, vesiculitis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis
Obstruction and cysts	Ejaculatory duct obstruction or calculi; seminal vesicle or utricle cyst
Tumours	**Prostate cancer**; rarely urethral, bladder, seminal vesicle
Vascular and systemic	Prostatic varices, hypertension, bleeding disorders, anticoagulants

Evaluation

- **History and examination:** biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants; blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, STI screen, **PSA (prostate-specific antigen)** over 40, coagulation if indicated.
- **Full work-up if:** age **≥40**, persistent or recurrent, haematuria, abnormal DRE or PSA, TB or cancer risk.
  - **TRUS (transrectal ultrasound):** seminal vesicles >1.5 cm or ejaculatory ducts >2.3 mm suggest obstruction.
  - **MRI (magnetic resonance imaging):** best for seminal vesicle haemorrhage, cysts, cancer.
  - **Cystoscopy and cytology**; early-morning urine for acid-fast bacilli.

Management

- Young, single episode, normal tests: **reassure**.
- Treat the cause: antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- Persistent prostatic bleeding: 5-ARI (5-alpha-reductase inhibitor).
- Refractory or obstructive: TURED (transurethral resection of ejaculatory ducts), cyst aspiration or unroofing, **seminal vesiculoscopy** (small scope through the ejaculatory duct or utricle; irrigation, laser for stones).

Diagram to draw: algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

Key points to remember:

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or with haematuria.
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

Q3e. Botulimum toxin

Type: SAQ (10 marks) (also answered in 2025-Winter Paper 2, Q3d)

"Botulimum" is read as botulinum toxin.

Introduction

- Neurotoxin of *Clostridium botulinum*, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**: a heavy chain (100 kDa) and a zinc-dependent light chain (50 kDa).

Mechanism

1. Heavy chain binds the cholinergic nerve terminal and the toxin is internalised.
2. Light chain **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE docking protein. Vesicles cannot fuse, so **acetylcholine release stops**.

- **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 and P2X3 sensory receptors, so urgency falls.
- Effect wanes as nerve terminals sprout (6–9 months), so injections are repeated.

Technique

- Cystoscopic detrusor injection, **20 sites (100 U) or 30 sites (200 U)**, usually sparing trigone; check PVR (post-void residual) at 2 weeks.

Urological uses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U**. ABC trial: equal to antimuscarinic, more complete dryness. ROSETTA: comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection)
NDO (neurogenic detrusor overactivity)	**200 U** (DIGNITY): less incontinence, larger capacity, lower pressure
Paediatric NDO (spina bifida)	About 10 U/kg (maximum 360 U) in published series; FDA (Food and Drug Administration) label 6 U/kg, maximum 200 U
DSD (detrusor–sphincter dyssynergia)	Intrasphincteric injection
IC/BPS (interstitial cystitis/bladder pain syndrome)	Off-label option
BPH (benign prostatic hyperplasia)	Intraprostatic: **no benefit** over placebo

Complications

- **Retention needing CISC (clean intermittent self-catheterisation)**, commoner in NDO; each 10 mL of baseline PVR raises risk about 27%. The patient must accept CISC beforehand.
- UTI, haematuria; rarely generalised weakness. Contraindications: active UTI, myasthenia gravis, pregnancy.

Diagram to draw: cholinergic synapse showing SNAP-25 cleavage by the light chain.

Key points to remember:

- Cleaves SNAP-25; efferent and afferent action.
- 100 U idiopathic OAB; 200 U NDO.
- Main risks: retention needing CISC, UTI.
- Children: FDA label 6 U/kg, maximum 200 U.

Q3f. Retrograde endopyelotomy

Type: SAQ (10 marks)

Definition and principle

- Endoscopic **full-thickness incision of the UPJ (ureteropelvic junction)** from inside the ureter, through to peripelvic fat, followed by stenting so the ureter heals with a wider lumen (Davis "intubated ureterotomy" concept, 1943).
- **Retrograde ureteroscopic holmium laser endopyelotomy** is now the standard endourological route: direct vision without a percutaneous tract.

Indications and contraindications

Favourable	Unfavourable / contraindicated
Functionally significant UPJO (UPJ obstruction); short segment	**Long stenotic segment**
Secondary UPJO (failed pyeloplasty)	**Upper-tract stones** (treat percutaneously instead)
Moderate hydronephrosis, fair function	**Crossing vessel**, **gross hydronephrosis**, poor function

- Pre-operative CT (computed tomography) angiography or Doppler to look for crossing vessels.

Technique

1. General anaesthesia; retrograde pyelogram; hydrophilic safety wire into the pelvis.
2. Access sheath; flexible ureteroscope to the UPJ (semirigid in some women). Dilate a narrow intramural ureter or pre-stent.
3. Holmium laser, 200-µm fibre, 0.8–1.2 J, 10–15 Hz.
4. **Lateral incision** (no crossing vessels laterally), deepened into fat and extended into normal ureter and pelvis.
5. Optional balloon dilation to 24 Fr; DJ (double-J) stent and Foley catheter.
6. Stent out after some weeks; **diuretic renogram 4 weeks later**; follow up for 24–36 months.

Results

- Holmium series 70–80% at 5 years (range 44–89%). Excluding crossing vessels >4 mm raises success to about 90%.
- Crossing vessel present versus absent: 42% versus 86% (antegrade series).
- Still below pyeloplasty (about 95%), but most cost-effective endourological option.

Complications

- Urine leak, stent migration, infection, bleeding (lateral cut spares vessels), stricture (rare with small scopes); ureteroarterial fistula reported.
- **Also know:** antegrade (percutaneous) endopyelotomy for UPJO with stones; cautery wire balloon (more haemorrhage); balloon dilation alone (about 42% success); failure then needs pyeloplasty.

Diagram to draw: UPJ with lateral full-thickness incision and crossing vessel lying anterior/posterior.

Key points to remember:

- Lateral full-thickness incision over a stent.
- Crossing vessel and gross hydronephrosis predict failure.
- Success 70–80%, below pyeloplasty (about 95%).

Rapid Revision Sheet – Practice Set 9 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Gaur 1992: glove-balloon retroperitoneoscopy.

Q3b. EMG (SAQ)

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

Q3c. Secondary malignancies in Uro (SAQ)

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